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REVIEW

Comparison of laser therapy and extracorporeal shock wave therapy in the treatment of patients with plantar fasciitis: a systematic review and meta-analysis of prospective studies

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Abstract

The purpose of this systematic review and meta-analysis was to evaluate and compare the effectiveness of LILT, HILT and ESWT for chronic PF. A thorough literature search across several databases yielded 108 studies. After screening for relevance, 10 prospective studies involving 803 patients were included in the analysis. Data was pooled using a random-effects model, with the mean difference (MD) and 95%CI as the main statistical measures. Our meta-analysis found no significant difference between ESWT and HILT in the change of VAS score and FFI score from pretreatment to post-treatment scores (MD: -0.50, 95% CI: -1.42 to 0.42, $P=0.28$), and (MD: -8.08, 95% CI: -22.09 to 5.94, $P=0.26$), respectively. Also, no difference between HILT and LILT in the change of VAS score from pretreatment to post-treatment scores (MD: 1.95, 95% CI: -0.84 to 4.74, $P=0.17$). Additionally, our analysis revealed no significant difference between ESWT and LILT in the change of AFOS score from pretreatment to post-treatment scores (MD: -1.64, 95% CI: -10.51 to 7.23, $P=0.72$). The analysis concluded that ESWT, LILT, and HILT all demonstrate similar efficacy in treating chronic plantar fasciitis, with no significant advantages observed for any specific therapy.

Keywords Laser therapy · Extracorporeal shock wave therapy · Visual analogue scale · Foot functional index · Plantar fasciitis

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Introduction

Plantar fasciitis (PF) is the most common cause of heel pain in adults, affecting approximately 10% of the adults with foot pain [1]. It is often linked to risk factors such as middle age, obesity, excessive foot pronation, and prolonged standing [2]. Traditionally considered an inflammatory condition, recent evidence suggests that PF is more accurately characterized as a degenerative disorder, or fasciopathy, reflecting chronic overuse and biomechanical stress [3, 4]. Patients typically present with pain during the first steps in the morning or after periods of inactivity, which may diminish with movement but can return or intensify later in the day [4].

The diagnosis of PF is usually clinical, supported by imaging techniques such as magnetic resonance imaging (MRI) and ultrasonography in complex cases [5]. Conservative treatment options, including rest, stretching exercises, nonsteroidal anti-inflammatory drugs (NSAIDs), physical therapy, and orthotic supports, provide relief for most patients [6, 7]. However, approximately 10% of cases remain resistant to these interventions, leading to the exploration of advanced therapies such as extracorporeal shockwave therapy (ESWT) and laser-based treatments [6].

ESWT, approved by the FDA for chronic PF, uses high-pressure sound waves to stimulate tissue repair by reactivating the chronic injury into an acute phase, thus promoting the natural healing process (8). Success rates for ESWT range from 34% to 88%, with minimal reported side effects, making it a safe and effective option for chronic PF management [8]. Meanwhile, low-level laser therapy (LLLT) and high-intensity laser therapy (HILT) have gained attention for their non-invasive nature and potential to reduce pain and inflammation through photobiomodulation and deeper tissue penetration, respectively [9].

Despite the growing use of these modalities, there is limited direct comparative research to guide clinicians in selecting the most effective treatment for chronic PF. This systematic review and meta-analysis aim to fill this gap by comparing the efficacy of ESWT, LLLT, and HILT in terms of pain reduction, functional improvement, and overall patient outcomes in chronic PF management.

Methods

Protocol registration and Documentation

The protocol for this systematic review and meta-analysis was carefully developed and registered with the International Prospective Register of Systematic Reviews (PROSPERO) under the ID: (CRD42025645907).

Search strategy

A comprehensive search was performed across medical databases, including Web of Science (WOS), Cochrane Library, and Medline (PubMed), Embase, Google Scholar and Scopus covering the period from inception to January 2025. Keywords relevant to the research objectives were identified, supplemented by terms from the MeSH database, and combined using Boolean operators "OR" and "AND." The complete search strategies tailored to each database are detailed in Table S1. Adjustments were made to accommodate the specific requirements of individual databases.

Study selection and eligibility criteria

The inclusion criteria for this review were as follows: (A) Randomized controlled trials (RCTs), prospective cohort, cross-sectional and case control studies. (B) Studies reporting quantitative and qualitative data on the efficacy and safety of laser therapy or Extracorporeal Shock Wave Therapy (ESWT) for managing plantar fasciitis. (C) Studies published in English. (D) No restrictions on patient age, sex, or country. The exclusion criteria included: (A) Papers published in languages other than English. (B) Pilot RCTs, RCT study protocols, retrospective studies, meta-analyses, systematic reviews, economic analyses, animal studies, narrative reviews, editorials, case series, and case reports.

Based on the predefined inclusion and exclusion criteria, a comprehensive search of all available literature was conducted from inception to January 2025 to identify relevant studies. A total of 108 articles were retrieved from the three databases. After filtering in Excel, a significant number of duplicates were removed. The screening process was conducted independently by two reviewers (AB and BC). The first stage involved screening titles and abstracts, while the second stage included full-text screening to exclude studies that did not meet the eligibility criteria or were irrelevant to the study. Disagreements at any stage were resolved by discussion; when consensus could not be reached, a third senior reviewer adjudicated.

Data Extraction, quality assessment, and evidence assessment

Data extracted from the included studies include first author name, year of publication, number of patients in each arm of the study, mean age, mean BMI, aim and conclusion of the study. Main outcomes were VAS score, FFI score and AFOS scale. Also, we extracted the inclusion and exclusion criteria of each included study.

Two reviewers independently performed the quality assessment, with any disagreements resolved through

discussion and, when necessary, adjudication by a third senior reviewer. We used the Cochrane Risk of Bias (ROB) assessment tool for RCTs, as outlined in the Cochrane Handbook for Systematic Reviews of Interventions, version 6.0. This tool identifies five types of bias: performance bias, selection bias, detection bias, reporting bias, and attrition bias. Based on these domains, each including RCT was categorized as having a high, unclear, or low risk of bias. ROBINS-I tool (Risk of Bias in Non-randomized Studies of Interventions), were used for non-randomized studies. ROBINS-I assesses potential bias across seven domains: confounding, selection of participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes, and selection of the reported result. Each domain was judged as having low, moderate, serious, or critical risk of bias, leading to an overall risk-of-bias rating for the study.

The GRADE approach (Grading of Recommendations, Assessment, Development, and Evaluations) was used to assess the quality of evidence. Randomized trials were initially rated as high quality but could be downgraded due to factors such as risk of bias, inconsistency, indirectness, imprecision, or other limitations. Conversely, evidence could be upgraded if a strong effect was observed or if confounding factors likely reduced the measured effect. Each outcome was ultimately assigned a quality rating of high, moderate, low, or very low.

Statistical analysis

Data analysis was conducted using Review Manager (RevMan) version 5.4.1. The extracted trial data were evaluated using mean difference (MD) and corresponding standard deviations, applying a random-effects model with the inverse variance statistical method. Key outcomes analyzed included VAS scores, FFI scores, and AFOS scales. Results were reported as MD with 95% confidence intervals (CI) and P-values, with statistical significance defined as $P<0.05$. Network meta-analysis random model effect was performed using R-studio software. Heterogeneity was assessed using the I^2 statistic. Subgroup analyses were performed to evaluate the efficacy and safety of treatments before and after intervention. The calculation of changes between pre- and post-treatment values was conducted using a specially designed calculation sheet. To ensure consistency, these calculations were independently performed by two authors.

Results

Search results

A comprehensive search across PubMed, Web of Science (WOS), Scopus, Embase, Google scholar and Cochrane databases initially identified 308 studies. Following the removal of duplicates, 228 unique studies remained for title and abstract screening, which resulted in the exclusion of 215 studies. 13 full-text articles were then independently reviewed, and 10 of them met the criteria for inclusion [1, 10–14]. [4, 15–17] the detailed search and selection process is illustrated in Fig. 1.

Summary of the included studies

This review includes a total of 10 studies involving 803 patients. Across these studies, the patient groups were distributed as follows: 245 patients received ESW, 228 received HILT, and 189 received LILT. Specifically, four studies compared LILT to ESW, three studies compared LILT to HILT, and three studies compared ESW to HILT. Most studies were randomized controlled trials (RCTs) conducted in Turkey, with additional studies from Pakistan, Iran, Thailand, Egypt, and Lithuania. Sample sizes varied between 30 and 102 participants, with mean ages ranging from 38.06 to 58.2 years. Mean BMI values were consistently reported in most studies, ranging between 23.69 and 33.1, except in two studies where BMI data were unavailable Table 1.

The included studies generally targeted adults aged 18 years or older, with some studies specifying an upper age limit (e.g., 55, 65, or 85 years). Common inclusion criteria were as follows: 1) Diagnosis of Chronic Plantar Fasciitis: Most studies required a confirmed diagnosis of plantar fasciitis, with symptoms persisting for at least 6 weeks to 6 months. 2)Visual Analog Scale (VAS) Pain Scores: Some studies included participants with moderate to severe pain, commonly defined as a VAS score$\geq$5. 3) Unresponsiveness to Conservative Treatment: Many participants had already failed to respond to conservative measures, such as NSAIDs, stretching, insoles, or physical therapy. 4) Specific Pain Characteristics: Heel pain at the first step in the morning, tenderness at the plantar fascia insertion site, and pain exacerbation with weight-bearing activities were commonly reported Table S2.

Exclusion criteria varied but frequently targeted factors that could confound results or pose risks, such as:

1) Previous Interventions: History of corticosteroid injections, foot or ankle surgery, or recent trauma were common reasons for exclusion.2) Systemic or Neurological Conditions: Patients with systemic diseases (e.g., rheumatoid arthritis, lupus, diabetes mellitus, or gout) or neurological impairments (e.g., radiculopathy or neuropathy) were excluded in most studies. 3) Acute or Infected Conditions: Acute plantar fasciitis, infections, open wounds, or observable skin changes at the treatment site were often excluded. 4) Contraindications

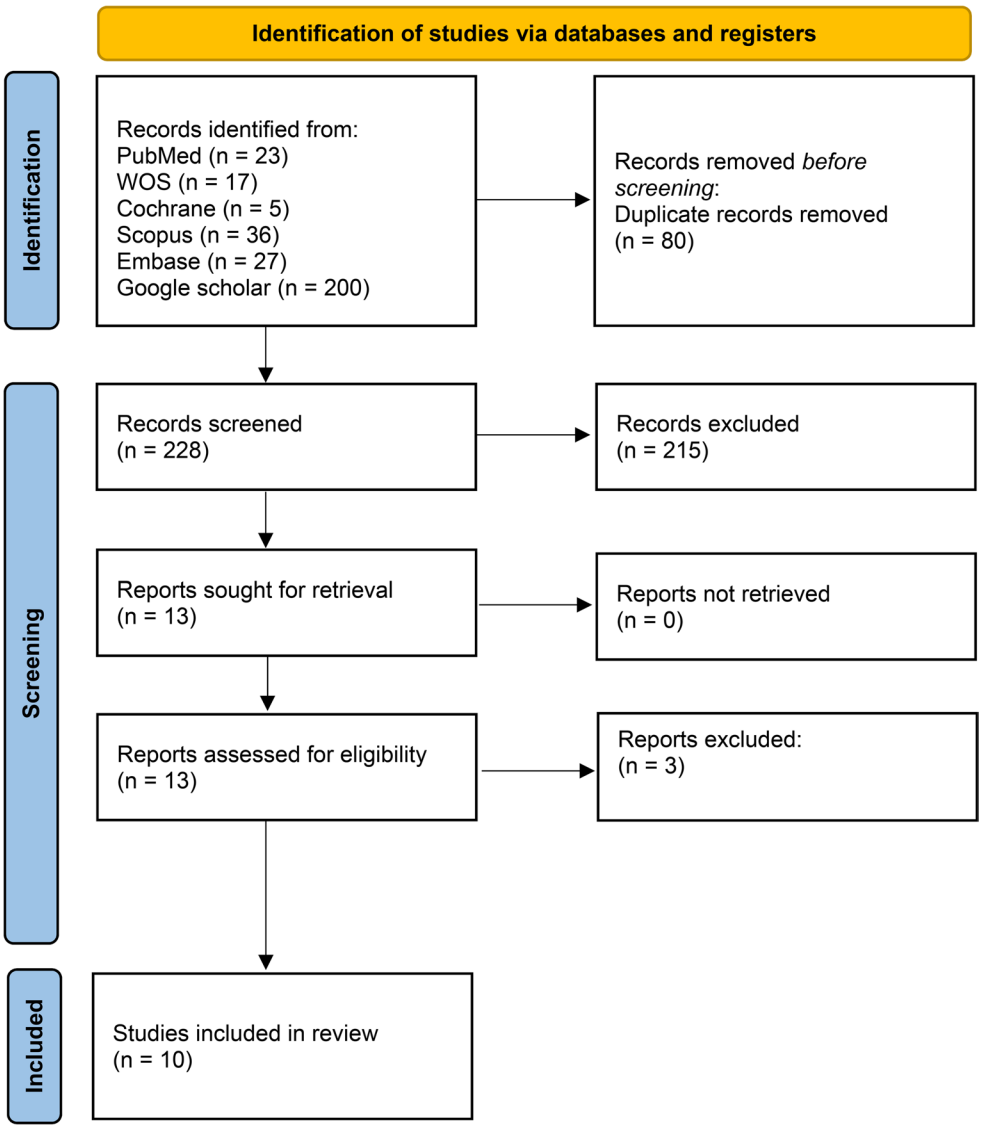


Fig. 1 Flow chart of the selection process

to Treatment: Pregnancy, pacemakers, metal implants, or conditions affecting healing (e.g., anticoagulant use, malignancy) were also common exclusion criteria.
5) BMI and Lifestyle Factors: Some studies excluded participants with high BMI (e.g., ≥ 30 or ≥40) or those using medications interfering with healing or pain perception Table S2.

Quality and GRADE assessment

We assessed the included RCTs using ROB-2 tool. Of 9 studies, 6 studies were considered to have low risk of bias (High quality). Only 3 studies were considered to have some concerns (moderate risk of bias) due to some concerns in deviation from the intended intervention and in the selection of the reported results. One study was assessed using ROBINS-I tools and was considered to have moderate risk of bias. All the details of the quality assessment process are provided in Figure S1 and Figure S2. According to GRADE the outcomes were of moderate and low quality Table S3.

Meta-analysis

ESWT versus HILT

Three studies assessed VAS. Our meta-analysis demonstrated no significant difference between ESWT and HILT in post-treatment scores (MD: −0.31, 95% CI: −1.48 to 0.85, $P=0.60$) or in the change from pretreatment to post-treatment scores (MD: −0.50, 95% CI: −1.42 to 0.42, $P=0.28$). The heterogeneity among studies was high for post-treatment analysis ($I^2 = 68\%$) but negligible for the change analysis ($I^2 = 0\%$). Figure 2 The funnel plot was unsymmetrical, indicating a presence of publication bias Figure S3.

Table 1 Characteristics of the included studies

Study	Type of the study	Country	Comparison	Number of patients	Mean age	Mean BMI	Aim	Conclusion
Riaz et al., 2023	RCT	Pakistan	ESW versus HILT	ESW (15), HILT (15)	ESW (39.66), HILT (38.06)	ESW (26.70), HILT (25.77)	To explore the best non-invasive treatment option for chronic PF within the scope of physical therapy.	Extracorporeal shockwave has been promising treatment for chronic plantar fasciitis but High intensity LASER therapy effectiveness was not explored a lot in previous literature. The study concluded that both ESWT and HILT are effective in reducing pain and improving function on VAS and FFI, though the ESWT group was found better than both groups in FFI score and VAS across time based on the mean differences across time.
Bidoki et al., 2024	RCT	Iran	ESW versus HILT	ESW (19), HILT (19)	ESW (45.05), HILT (44.26)	ESW (26.92), HILT (27.47)	This study aimed to compare High-intensity Laser Therapy (HILT) with Extracorporeal Shock Wave Therapy (ESWT) in patients with PF.	ESWT and HILT decrease pain and increase patient satisfaction in PF. Besides, both methods are non-invasive and safe. However, there is a significant difference between them, and HILT is more effective.
Thammajaree et al., 2023	RCT	Thailand	ESW versus HILT	ESW (16), HILT (16)	ESW (48.12), HILT (46.06)	ESW (24.00), HILT (23.69)	This study aimed to compare the effects of radial extracorporeal shockwave therapy (rESWT) to the effects of high intensity laser therapy (HILT) in the treatment of individuals with plantar fasciitis.	This study proved that either rESWT or HILT can success fully reduce all three aspects of pain within six sessions of the interventions, no intervention was superior to the other regarding pain management. However, HILT seemed to be better in reducing FFI (functional limitation) after six sessions of the intervention program. It was found that either rESWT or HILT can alleviate pain and other clinical variables.
Timurtas et al., 2024	RCT	Turkey	ESW versus LILT	ESW (27), LILT (20)	ESW (46.9), LILT (46.8)	ESW (31.6), LILT (33.1)	We compared the short-term effectiveness of ESWT and LLLT on pain and function in patients with plantar fasciitis.	Compared with ESWT, LLLT was found to be superior as an effective approach in the short-term management of plantar fasciitis.
Koz et al., 2023	Prospective cross-sectional study	Turkey	ESW versus LILT	ESW (22), LILT (18)	ESW (49,23), LILT (49,28)	ESW (31.4), LILT (32.4)	In this study it was aimed to compare efficacies of Extracorporeal Shockwave Therapy (ESWT) and Low-Level Laser Therapy (LLLT) on patients with PF.	The results of this study indicated significant improvements in terms of pain, functional status and daily life activities following the administration of either of the treatments. Furthermore, LLLT was found to be significantly more effective for alleviating pain than ESWT in the treatment of PF
ELsehrawy et al., 2018	RCT	Egypt	ESW versus LILT	ESW (23), LILT (23)	ESW (46.00), LILT (46.4)	ESW (29.5), LILT (32.8)	To compare and evaluate the effectiveness of extracorporeal shock wave therapy (ESWT) and low-level laser therapy (LLLT) using diagnostic ultrasound (US) in the management of chronic plantar fasciitis (PF).	LILT proved to be more effective than ESWT in pain relief in patients with PF after one month of treatment.

Table 1 (continued)

Study	Type of the study	Country	Comparison	Number of patients	Mean age	Mean BMI	Aim	Conclusion
Ulusoy et al., 2017	RCT	Turkey	ESW versus LILT	ESW (20), LILT (20)	ESW (53.4), LILT (54.45)	ESW (31.94), LILT (32.01)	We determined and compared the effectiveness of low-level laser therapy (LLLT), therapeutic ultrasound (US) therapy, and extracorporeal shock wave therapy (ESWT) using magnetic resonance imaging (MRI).	The treatment of chronic plantar fasciitis with LLLT and ESWT resulted in similar outcomes, and both were more successful than US therapy in pain improvement and functional outcomes.
Sanmak et al., 2019	RCT	Turkey	ESW versus LILT	ESW (17), LILT (17)	ESW (49), LILT (53)	/	The aim of this study is to compare the efficacy of extracorporeal shock wave therapy (ESWT) and low-level laser therapy (LLLT) in terms of fascia thickness, heel pain, and foot functions in patients with plantar fasciitis (PF).	Our study results suggest that both ESWT and LLLT seem to be effective on pain, foot functions, and fascia thickness in the treatment of PF.
Ordahan et al., 2018	RCT	Turkey	HILT versus LILT	HILT (35), LILT (35)	HILT (48.73), LILT (48.65)	HILT (31.16), LILT (31.22)	Weaimed to compare the efficacy of low-level laser therapy (LLLT) and high-intensity laser therapy (HILT) in the treatment of plantar fasciitis (PF).	The HILT group demonstrated betterimprovementinall parameters than the LLLT group. Although both treatments improved the pain levels, function, and quality of life in patients with PF, HILT had a more significant effect than LLLT
Narusevicuite et al., 2020	RCT	Lithuania	HILT versus LILT	HILT (51), LILT (51)	HILT (54.2), LILT (58.2)	/	To evaluate and compare the efficacy of high-intensity laser therapy (HILT) and low-level laser therapy (LLLT) for plantar fasciitis.	No statistically significant difference between groups was observed.

Two studies assessed the FFI score. Our meta-analysis demonstrated no significant difference between ESWT and HILT in post-treatment scores (MD: 3.16, 95% CI: −33.27 to 39.59, $P=0.86$) or in the change from pretreatment to post-treatment scores (MD: −8.08, 95% CI: −22.09 to 5.94, $P=0.26$). The heterogeneity among studies was high for post-treatment analysis (I^2 = 87%) but negligible for the change analysis (I^2 = 0%). Figure 3 The funnel plot was unsymmetrical, indicating a presence of publication bias Figure S4.

LILT versus HILT

Two studies assessed VAS. Our meta-analysis demonstrated no significant difference between LILT and HILT in post-treatment scores (MD: 1.29, 95% CI: −1.77 to 4.34, $P=0.41$) or in the change from pretreatment to post-treatment scores (MD: 1.95, 95% CI: −0.84 to 4.74, $P=0.17$). The heterogeneity among studies was high for post-treatment analysis (I^2 = 93%) and pretreatment to post-treatment analysis (I^2 = 88%). A potential risk of bias was identified through the pretreatment analysis which showed a significantly lower VAS score in LILT group than HILT group (MD: −0.64, 95% CI: −1.22 to −0.06, $P=0.03$). Figure 4 The funnel plot was unsymmetrical, indicating a presence of publication bias Figure S5.

ESWT versus LILT

Four studies assessed VAS. Our meta-analysis demonstrated no significant difference between LILT and ESWT in post-treatment scores (MD: 0.47, 95% CI: −0.29 to 1.24, $P=0.22$) or in the change from pretreatment to post-treatment scores (MD: 0.70, 95% CI: −0.23 to 1.64, $P=0.14$). The heterogeneity among studies was low for post-treatment analysis (I^2 = 33%) and pretreatment to post-treatment analysis (I^2 = 22%). Figure 5 The funnel plot was unsymmetrical, indicating a presence of publication bias Figure S6.

Three studies assessed the FFI score. Our meta-analysis demonstrated no significant difference between ESWT and LILT in post-treatment scores (MD: 3.17, 95% CI: −8.38 to 14.72, $P=0.59$) or in the change from pretreatment to post-treatment scores (MD: 0.28, 95% CI: −13.32 to 13.88, $P=0.97$). The heterogeneity among

Table 2 GRADE assessment

Certainty assessment							№ of patients		Effect		Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	-	-	Relative (95% CI)	Absolute (95% CI)	
VAS score to compare ESWT and HILT.											
2	randomised trials	not serious	Serious[a]	not serious	Serious[b]	none	50	50	-	MD **−0.31 MD lower** (−1.48 lower to 0.85 higher)	⨁⨁◯◯ LOW
FFI score to compare ESWT and HILT.											
2	randomised trials	not serious	Serious[a]	not serious	Serious[b]	none	31	31	-	MD **3.16 MD lower** (−33.27 lower to 39.59 higher)	⨁⨁◯◯ LOW
VAS score to compare LILT and HILT.											
2	randomised trials	not serious	Serious[a]	not serious	Serious[b]	none	86	86	-	MD **1.29 MD lower** (−1.77 lower to 4.34 higher)	⨁⨁◯◯ LOW
VAS score to compare LILT and ESWT.											
4	randomised trials	not serious	not serious	not serious	Serious[b]	none	82	75	-	MD **0.47 MD lower** (−0.29 lower to 1.24 higher)	⨁⨁⨁◯ MODERATE
FFI score to compare ESWT and LILT.											
3	randomised trials	not serious	Serious[a]	not serious	Serious[b]	none	62	58	-	MD 3.17 **MD higher** (−8.38 lower to 14.72 higher)	⨁⨁◯◯ LOW
AFOS score to compare ESWT and LILT.											
2	randomised trials	not serious	not serious	not serious	Serious[b]	none	42	35		MD 0.18 **MD lower** (−5.69 lower to 6.04 higher)	⨁⨁⨁◯ MODERATE

Explanations: (a) Substantial heterogeneity, (b) CI crossed the line of null effect

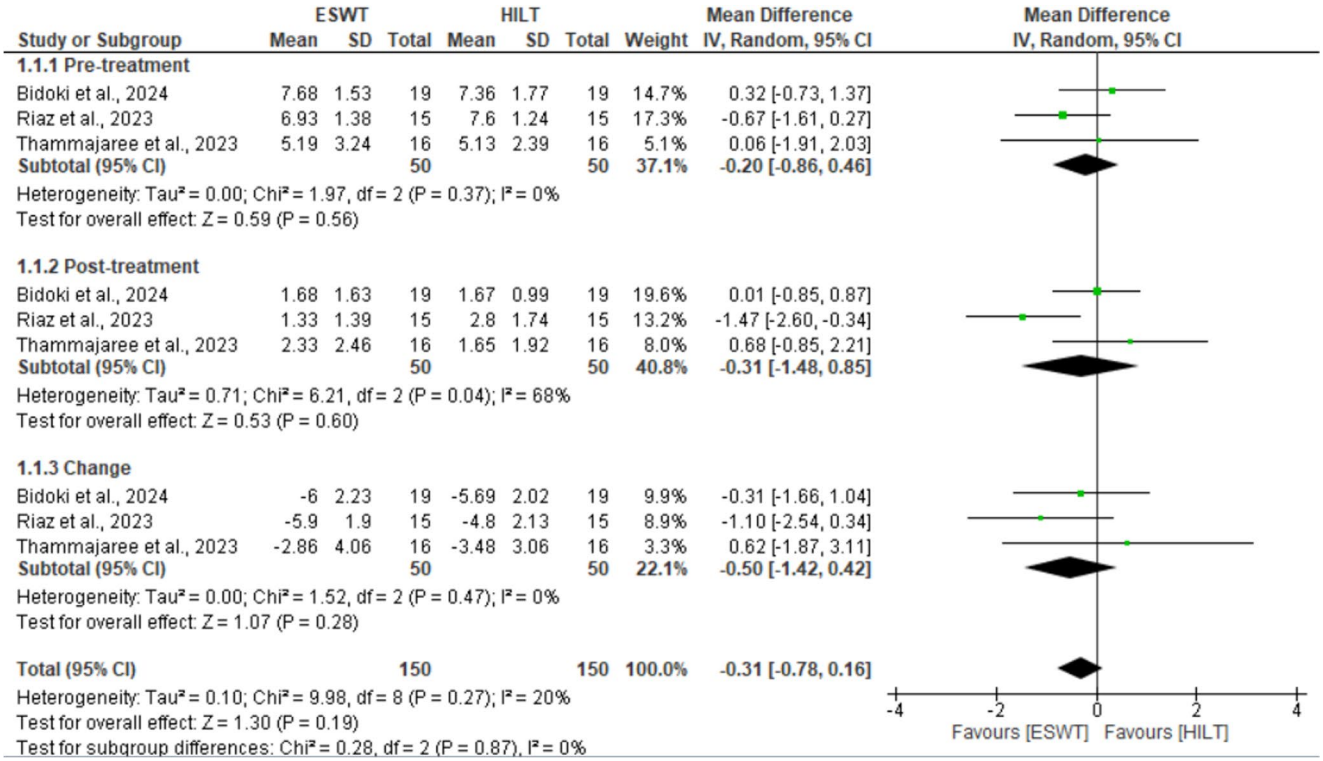


Fig. 2 Meta analysis of VAS score to compare ESWT and HILT

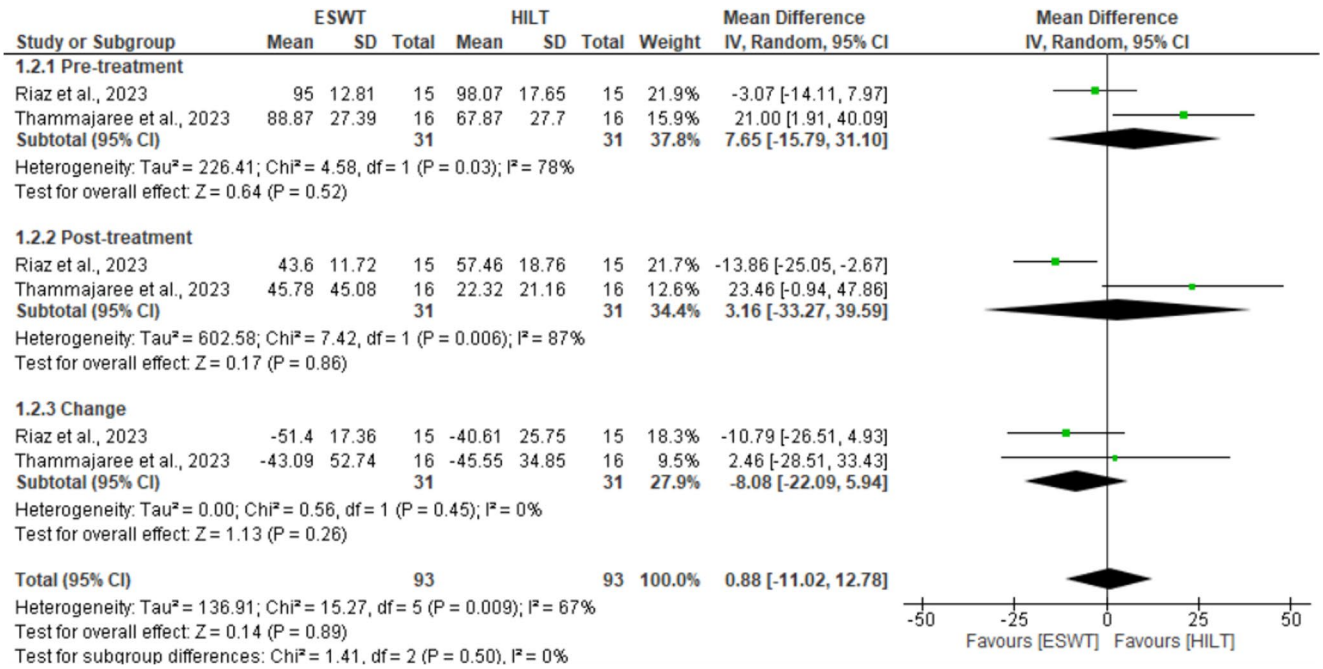


Fig. 3 Meta analysis of FFI score to compare ESWT and HILT

studies was high for post-treatment analysis (I^2 = 65%) and pretreatment to post-treatment analysis (I^2 = 53%). Figure 6 The funnel plot was unsymmetrical, indicating a presence of publication bias Figure S7.

Two studies assessed the American Orthopedic Foot and Ankle Society (AFOS) score. Our meta-analysis demonstrated no significant difference between ESWT and LILT in post-treatment scores (MD: 0.18, 95% CI: −5.69 to 6.04,

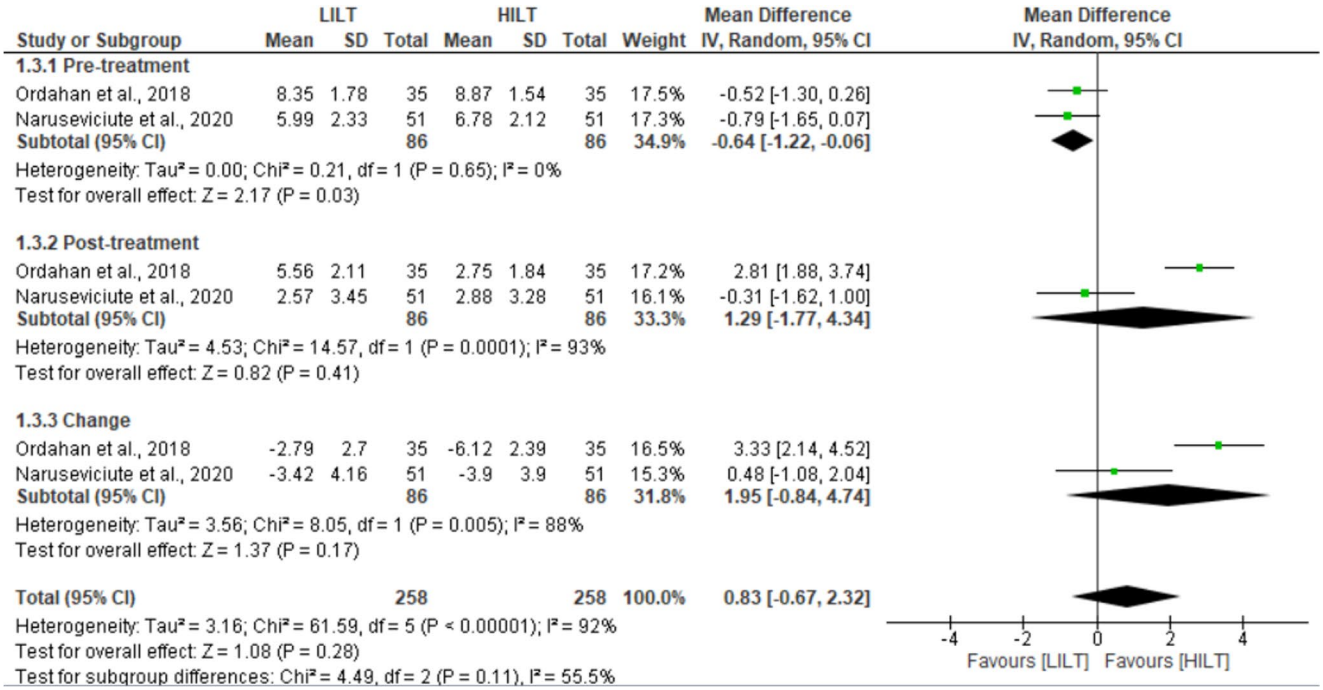


Fig. 4 Meta analysis of VAS score to compare LILT and HILT

P=0.95) or in the change from pretreatment to post-treatment scores (MD: −1.64, 95% CI: −10.51 to 7.23, *P*=0.72). The heterogeneity among studies was zero for post-treatment analysis (I^2 = 0%) and pretreatment to post-treatment analysis (I^2 = 0%). Figure 7 The funnel plot was unsymmetrical, indicating a presence of publication bias Figure S8.

Network meta-analysis

None of the treatments (ESWT, HILT, or LILT) showed a statistically significant improvement in VAS scores Figure S9 & Figure S10. HILT in the network meta-analysis (−1.97, 95%CI: −4.33, 0.39) suggests the largest potential improvement, but still lacks statistical significance Table S3.

Discussion

This systematic review and meta-analysis aimed to compare the efficacy of ESWT, LILT, and HILT in managing chronic PF. The analysis revealed no statistically significant differences between these treatment modalities regarding VAS scores, FFI scores, or AFOS. While each intervention showed clinical improvements over time, the absence of between-group differences indicates that these modalities are comparably effective in addressing the symptoms of chronic PF.

Comparing the physiological mechanisms of the three modalities provides additional insight into their effectiveness. ESWT induces neovascularization, reduces inflammatory mediators, and promotes tissue regeneration through acoustic mechanical waves [18]. HILT, on the other hand, enhances ATP synthesis and tissue healing through photothermal effects, while LILT modulates inflammation and stimulates cell repair via photochemical mechanisms [19, 20]. Despite these distinct mechanisms, all three therapies target pain reduction and functional improvement, which likely explains the comparable outcomes observed in this analysis.

The findings of this study align with existing literature, which highlights the therapeutic potential of ESWT, LILT, and HILT in managing chronic PF. For instance, HILT has been reported as a promising treatment option for other musculoskeletal conditions, such as trapezius myofascial pain syndrome, Achilles tendinopathy, and knee osteoarthritis [21–23]. Previous studies have noted that HILT may outperform LILT in certain contexts due to its deeper penetration and photothermal effects, which promote tissue healing and reduce inflammation [24]. However, the current analysis did not find HILT to be significantly more effective than ESWT or LILT for PF management. This difference may stem from variations in treatment protocols and patient populations.

ESWT, a widely recognized non-invasive treatment for chronic PF, has demonstrated significant long-term benefits in several studies, with improvements sustained for up to 12 months post-treatment. For example, Gollwitzer et al. [25] and Dastgir et al. [27] reported substantial pain relief and functional improvements with ESWT. Similar outcomes

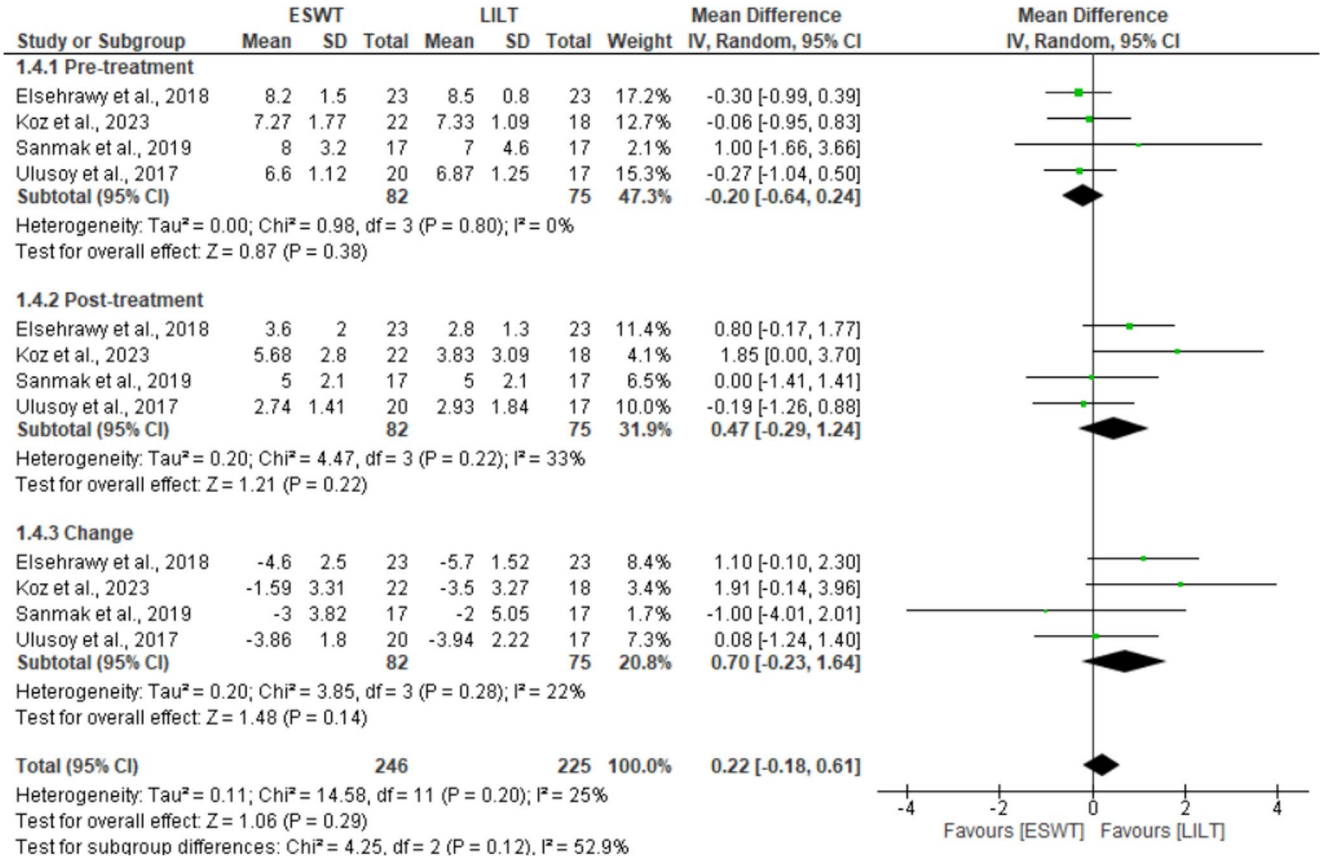


Fig. 5 Meta analysis of VAS score to compare LILT and ESWT

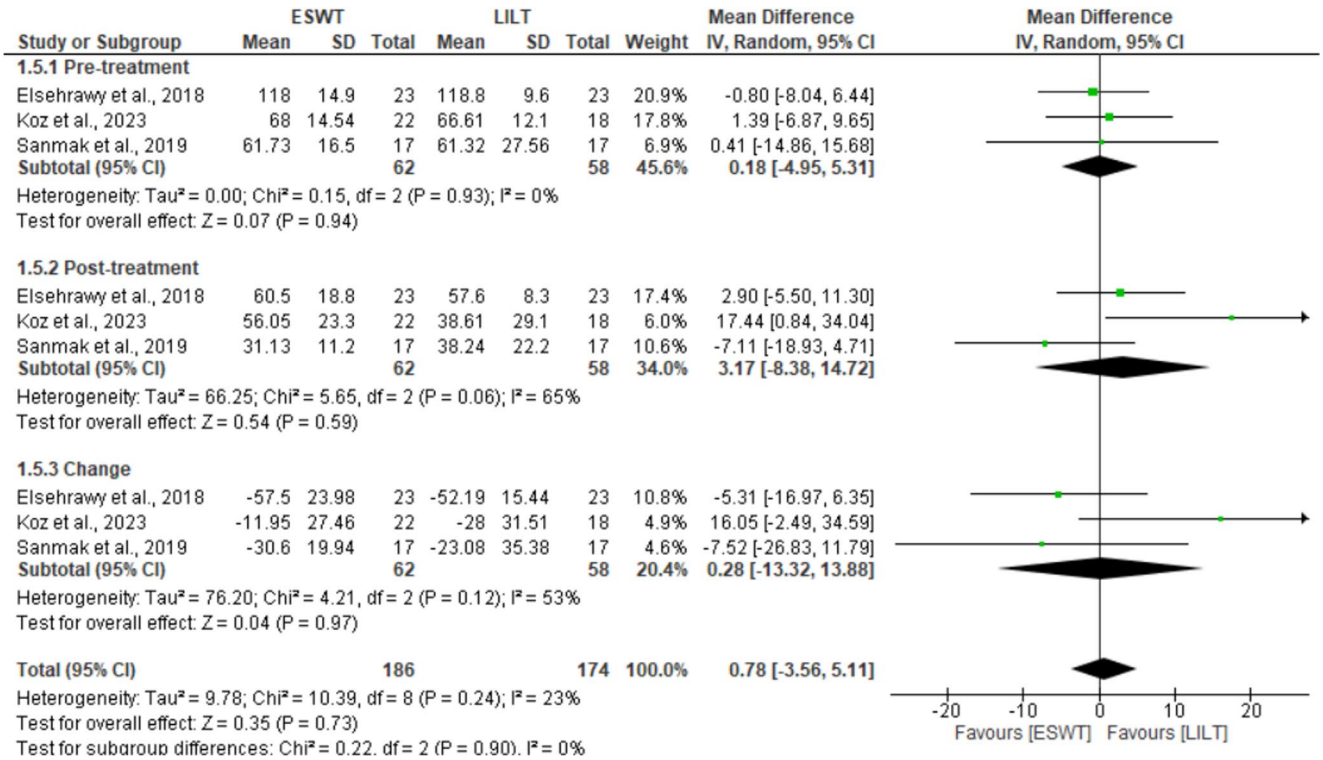


Fig. 6 Meta analysis of FFI score to compare ESWT and LILT

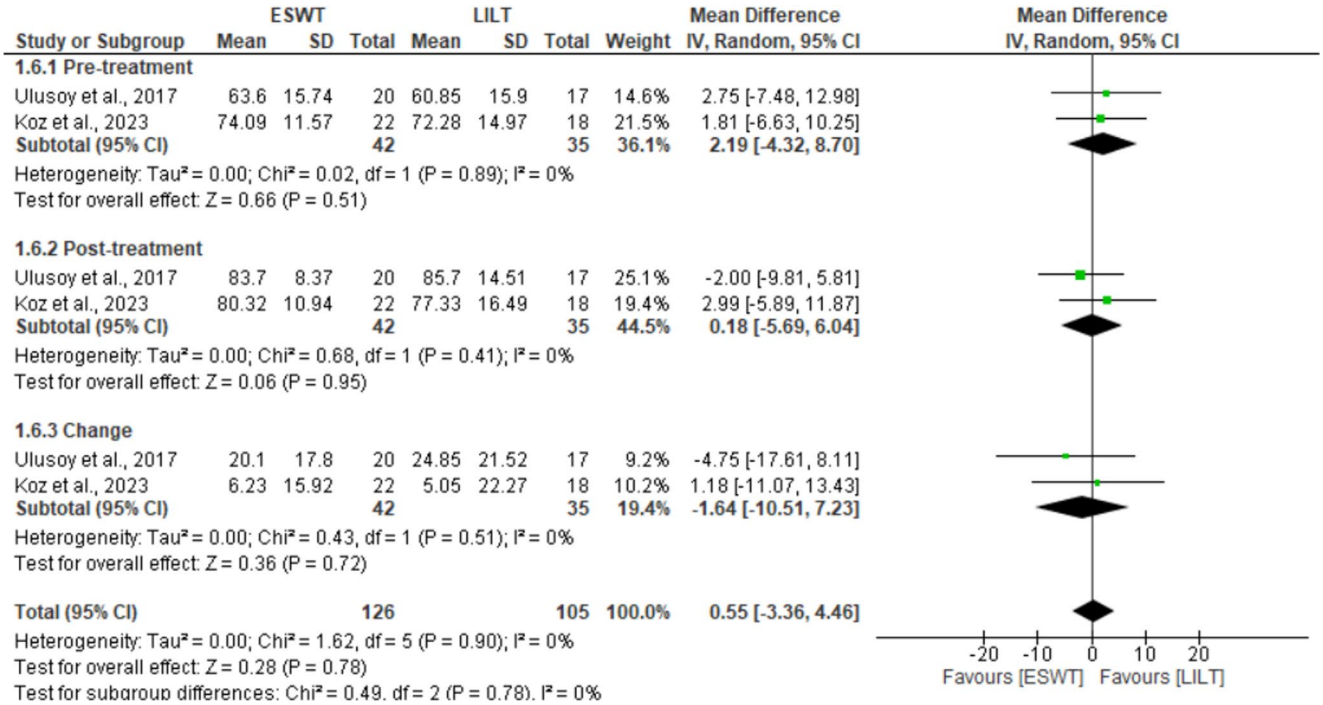


Fig. 7 Meta analysis of AFOS score to compare ESWT and LILT

were observed in studies comparing ESWT to placebo [26]. Despite these findings, the present meta-analysis did not establish ESWT as superior to HILT or LILT. These results are consistent with some studies that reported equivalent outcomes between ESWT and other modalities [27]. Variability in shock wave energy levels, focus zones, and patient adherence to treatment protocols could partially explain the differences observed across studies.

LILT has gained popularity as an effective option for short-term symptom relief in PF. Study by Kiritsi et al. [28] and Macias et al. [29] demonstrated that LILT significantly reduces pain and improves plantar fascia thickness. This modality's ability to provide rapid symptom alleviation makes it a compelling option for patients seeking immediate relief. Additionally, combining LILT with home exercise programs has been shown to enhance treatment outcomes, as supported by previous studies [30, 31]. However, the current analysis found no significant advantage of LILT over ESWT or HILT, suggesting that its efficacy might be comparable rather than superior.

Several included studies highlighted the potential for combining these therapies with other treatment approaches. For instance, ESWT paired with home exercise therapy has been shown to improve foot performance and reduce pain more effectively than ESWT alone [32]. Similarly, combining HILT with traditional physiotherapy may yield better outcomes for PF and other musculoskeletal conditions [7]. The results of this meta-analysis emphasize the importance of personalized treatment approaches. Although ESWT, LILT, and HILT all demonstrated clinical benefits, no single intervention emerged as superior. Clinicians should consider patient preferences, treatment accessibility, and cost-effectiveness when selecting a modality. Furthermore, combining these therapies with traditional physiotherapy or home exercise programs may enhance outcomes, particularly in chronic cases where isolated treatments may have limited effects [30].

Several limitations must be acknowledged. First, the heterogeneity in treatment protocols, including variations in laser dosages, shock wave intensities, and treatment frequencies, may have influenced the results. Differences in pulse energy or session duration could account for the variability in outcomes across studies. Second, the lack of ultrasonographic evaluations in some included studies limited the ability to objectively assess changes in plantar fascia thickness, a key indicator of treatment efficacy. Third, the short follow-up periods in certain studies may underestimate the long-term effectiveness of these modalities. Fourth, this review includes only English-language publications, which may have introduced language bias. Additionally, some heterogeneity was observed in the reported outcomes, which may have affected the consistency of the findings.

Conclusion

In conclusion, this systematic review and meta-analysis provide robust evidence that ESWT, LILT, and HILT are effective options for managing chronic PF, with no single modality demonstrating clear superiority. These findings highlight the importance of individualized treatment plans tailored to patient needs and preferences. While all three modalities offer significant clinical benefits, further research is necessary to refine treatment protocols, evaluate long-term outcomes, and investigate the potential advantages of combined therapy approaches.

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Declarations

This study conducted under PRISMA guidelines and in accordance with the Declaration of Helsinki.

Human Ethics and Consent to Participate Not applicable.

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References

1. Ulusoy A, Cerrahoglu L, Orguc S (2017) Magnetic resonance imaging and clinical outcomes of laser therapy, ultrasound therapy, and extracorporeal shock wave therapy for treatment of plantar fasciitis: a randomized controlled trial. J Foot Ankle Surg 56:762–767. https://doi.org/10.1053/j.jfas.2017.02.013
2. Riddle DL, Pulisic M, Pidcoe P, Johnson RE (2003) Risk factors for plantar fasciitis. J Bone Joint Surg Am 85:872–877. https://doi.org/10.2106/00004623-200305000-00015
3. Dizon JNC, Gonzalez-Suarez C, Zamora MTG, Gambito EDV (2013) Effectiveness of extracorporeal shock wave therapy in chronic plantar fasciitis. Am J Phys Med Rehabil 92:606–620. https://doi.org/10.1097/PHM.0b013e31828cd42b
4. Karahan A, Ordahan B, Ozkuk K (2018) The effect of paraffin bath therapy versus extracorporeal shockwave therapy in the management of plantar fasciitis: a randomized clinical trial. Ann Phys Rehabil Med 61:e149. https://doi.org/10.1016/j.rehab.2018.05.334
5. League AC (2008) Current concepts review: plantar fasciitis. Foot Ankle Int 29:358–366. https://doi.org/10.3113/FAI.2008.0358
6. Thomas JL, Christensen JC, Kravitz SR et al (2010) The diagnosis and treatment of heel pain: a clinical practice guideline–revision 2010. J Foot Ankle Surg 49:S1–S19. https://doi.org/10.1053/j.jfas.2010.01.001
7. AssadS,Ahmad A, Kiani I et al (2016) Novel and Conservative approaches towards effective management of plantar fasciitis. Cureus https://doi.org/10.7759/cureus.913
8. Wang C-J (2012) Extracorporeal shockwave therapy in musculoskeletal disorders. J Orthop Surg Res 7:11. https://doi.org/10.1186/1749-799X-7-11
9. Rola P, Doroszko A, Derkacz A (2014) The use of low-level energy laser radiation in basic and clinical research. Adv Clin Exp Med 23:835–842. https://doi.org/10.17219/acem/37263
10. Timurtaş E, Çinar E, Selçuk H et al (2024) Extracorporeal shockwave therapy versus Low-Level laser therapy in the treatment of plantar fasciitis: A randomized controlled trial. J Am Podiatr Med Assoc 114. https://doi.org/10.7547/22-095
11. Yinilmez Sanmak ÖD, Külcü DG, Mesci N, Altunok EÇ (2019) Comparison of effects of low-level laser therapy and extracorporeal shock wave therapy in plantar fasciitis treatment: a randomized, prospective, single-blind clinical study. Turk J Phys Med Rehabil 65:184–190. https://doi.org/10.5606/tftrd.2019.3528
12. Koz G, Kamanli A, Kaban N, Harman H (2023) Efficacies of extracorporeal shockwave therapy and low-level laser therapy in patients with plantar fasciitis. Foot Ankle Surg 29:223–227. https://doi.org/10.1016/j.fas.2023.01.009
13. Naruseviciute D, Kubilius R (2020) The effect of high-intensity versus low-level laser therapy in the management of plantar fasciitis: randomized participant blind controlled trial. Clin Rehabil 34:1072–1082. https://doi.org/10.1177/0269215520929073
14. ELsehrawy G, Nasef S, Ibrahim M, Omar A (2018) Extracorporeal shock wave therapy versus low-level laser therapy in the management of chronic plantar fasciitis. Suez Canal University Medical Journal 21:71–81. https://doi.org/10.21608/scumj.2018.42935
15. Thammajaree C, Theapthong M, Palee P et al (2023) Effects of radial extracorporeal shockwave therapy versus high intensity laser therapy in individuals with plantar fasciitis: A randomised clinical trial. Lasers Med Sci 38:12–17. https://doi.org/10.1007/s10103-023-03791-5

16. Zare Bidoki M, Vafaeei Nasab MR, Khatibi Aghda A (2024) Comparison of high-intensity laser therapy with extracorporeal shock wave therapy in the treatment of patients with plantar fasciitis: a double-blind randomized clinical trial. Iran J Med Sci 49:147–155. https://doi.org/10.30476/ijms.2023.98042.2991
17. Riaz S, Sattar A, Seemal P et al (2023) Comparison of extracorporeal shockwave and high-intensity laser in treating chronic plantar fasciitis. Pakistan Journal of Medical and Health Sciences 17:46–49. https://doi.org/10.53350/pjmhs202317546
18. Wang C, Wang F, Yang KD et al (2003) Shock wave therapy induces neovascularization at the tendon–bone junction. A study in rabbits. J Orthop Res 21:984–989. https://doi.org/10.1016/S0736-0266(03)00104-9
19. Ezzati K, Laakso E-L, Salari A et al (2020) The beneficial effects of high-intensity laser therapy and co-interventions on musculoskeletal pain management: a systematic review. J Lasers Med Sci 11:81–90. https://doi.org/10.15171/jlms.2020.14
20. Silveira PCL, Scheffer D, Glaser V et al (2016) Low-level laser therapy attenuates the acute inflammatory response induced by muscle traumatic injury. Free Radic Res 50:503–513. https://doi.org/10.3109/10715762.2016.1147649
21. Akaltun MS, Altindag O, Turan N et al (2021) Efficacy of high intensity laser therapy in knee osteoarthritis: a double-blind controlled randomized study. Clin Rheumatol 40:1989–1995. https://doi.org/10.1007/s10067-020-05469-7
22. Dundar U, Turkmen U, Toktas H et al (2015) Effect of high-intensity laser therapy in the management of myofascial pain syndrome of the trapezius: a double-blind, placebo-controlled study. Lasers Med Sci 30:325–332. https://doi.org/10.1007/s10103-014-1671-8
23. Abdelkader NA, Helmy MNK, Fayaz NA, Saweeres ESB (2021) Short- and intermediate-term results of extracorporeal shockwave therapy for noninsertional Achilles tendinopathy. Foot Ankle Int 42:788–797. https://doi.org/10.1177/1071100720982613
24. Saleh MS, Shahien M, Mortada H et al (2024) High-intensity versus low-level laser in musculoskeletal disorders. Lasers Med Sci 39:179. https://doi.org/10.1007/s10103-024-04111-1
25. Gollwitzer H, Diehl P, von Korff A et al (2007) Extracorporeal shock wave therapy for chronic painful heel syndrome: a prospective, double blind, randomized trial assessing the efficacy of a new electromagnetic shock wave device. J Foot Ankle Surg 46:348–357. https://doi.org/10.1053/j.jfas.2007.05.011
26. Aqil A, Siddiqui MRS, Solan M et al (2013) Extracorporeal shock wave therapy is effective in treating chronic plantar fasciitis: a meta-analysis of RCTs. Clin Orthop Relat Res 471:3645–3652. https://doi.org/10.1007/s11999-013-3132-2
27. Dastgir N (2014) Extracorporeal shock wave therapy for treatment of plantar fasciitis. J Pak Med Assoc 64:675–678
28. Kiritsi O, Tsitas K, Malliaropoulos N, Mikroulis G (2010) Ultrasonographic evaluation of plantar fasciitis after low-level laser therapy: results of a double-blind, randomized, placebo-controlled trial. Lasers Med Sci 25:275–281. https://doi.org/10.1007/s10103-009-0737-5
29. Macias DM, Coughlin MJ, Zang K et al (2015) Low-level laser therapy at 635 nm for treatment of chronic plantar fasciitis: a placebo-controlled, randomized study. J Foot Ankle Surg 54:768–772. https://doi.org/10.1053/j.jfas.2014.12.014
30. Alfredo PP, Bjordal JM, Junior WS et al (2021) Efficacy of low-level laser therapy combined with exercise for subacromial impingement syndrome: a randomised controlled trial. Clin Rehabil 35:851–860. https://doi.org/10.1177/0269215520980984
31. Germano Maciel D, da Trajano Silva M, Rodrigues JA et al (2018) Low-level laser therapy combined to functional exercise on treatment of fibromyalgia: a double-blind randomized clinical trial. Lasers Med Sci 33:1949–1959. https://doi.org/10.1007/s10103-018-2561-2
32. Burton I (2022) Combined extracorporeal shockwave therapy and exercise for the treatment of tendinopathy: a narrative review. Sports Med Health Sci 4:8–17. https://doi.org/10.1016/j.smhs.2021.11.002

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